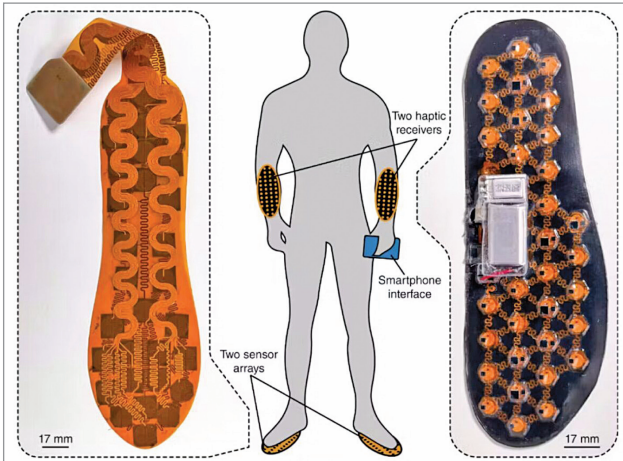


Wearable system helps restore foot and arm sensation

Researchers at the Georgia Institute of Technology have developed a wearable system that restores foot-pressure awareness in people with sensory impairments using pressure-sensing



Sensations from the feet are transmitted to the forearms (Photo Credit: Georgia Institute of Technology)

insoles and forearm-mounted haptic devices. The wireless system converts foot-pressure data into vibration and heat patterns on the forearms, improving balance and walking in stroke and spinal cord injury trials after brief training. Researchers also see potential for preventing diabetic foot ulcers and pressure injuries through thermal feedback.

Smart bicycle can tell when riders mean to turn

Researchers at the Shibaura Institute of Technology have developed an AI-based rider-assistance system that distinguishes intentional turning from loss of balance, activating stability support only when needed. Developed in Japan, the technology was validated on a steer-by-wire bicycle using a long short-term memory neural network trained with K-



The steer-by-wire bicycle platform developed in this study uses haptic feedback and a machine-learning-based control framework to estimate rider intent and distinguish intentional turning maneuvers from unintended instability (Credit: Associate Prof. Hiroaki Kuwahara from Shibaura Institute of Technology, Japan)

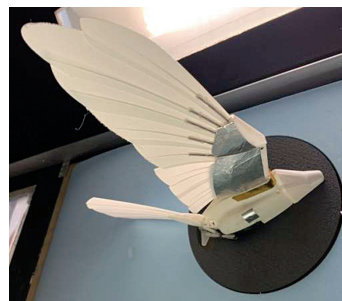
means clustering. By analysing steering angle, speed, roll angle, lateral acceleration, and steering torque in real time, the system avoided unnecessary intervention during normal cornering while assisting during instability. The approach could improve safety in electric bicycles, motorcycles, shared mobility fleets, and delivery vehicles.

Radiation-hardened 45nm space chip passes strategic mission tests

BAE Systems has completed strategic radiation-hardness testing of its Endura 45nm radiation-hardened system-on-chip (SoC), validating reliable operation in harsh space radiation environments. Built on a 45nm silicon-on-insulator process, the processor is designed for satellites and long-duration spacecraft requiring dependable, low-power onboard computing. It resists both single-event upsets and cumulative radiation damage while integrating secure boot, networking, L1/L2 cache, and FPGA functionality for mission-specific acceleration. The space-grade SoC is intended for military, scientific, and commercial spacecraft, with development units now available for software development and system integration in future missions.

Robotic bird advances drone flight

Researchers from RMIT University and the University of Bristol have developed a bio-inspired robotic bird that reveals how kestrels maintain stable flight in turbulent conditions, offering new strategies for resilient drone de-



The robotic kestrel in RMIT's industrial wind tunnel (Credit: RMIT University)

sign. Developed through experiments conducted at RMIT in Australia, the robotic platform replicates a nankeen kestrel's wing, tail, and body movements using motion-capture data from live birds. The study showed that coordinated wing and tail morphing, rapid

body adjustments, and flexible feathers improve stability in gusty air. The findings could enhance flight-control systems for small UAVs used in inspection, agriculture, search-and-rescue, and delivery applications.

Laser technique speeds up nanomaterial production

The University of Hyderabad and the Indian Institute of Technology Hyderabad have developed a laser-based technique to produce transition metal dichalcogenide (TMDC) nanosheets for electronics, photonics, and sensing applications. The method uses a femtosecond laser with a Bessel beam profile to separate single- or few-layer nanosheets from a synthesised TMDC pellet by breaking the weak forces between its layers without changing the material's chemical structure. Unlike mechanical exfoliation, chemical exfoliation, and chemical vapour deposition, the new approach produces single or multilayer nanosheets in less time with fewer processing steps.